R13

Code No: 111AD

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, December - 2017
ENGINEERING PHYSICS
(Common to CE, EEE, ME, ECE, CSE, CHEM, EIE, IT, MCT, ETM, MMT, AE, AME, MIE, PTM, MSNT, AGE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a)	What is unit cell?	[2]
b)	What are the Miller indices? How they are obtained?	[3]
c)	Define de-Broglie's hypothesis.	[2]
d)	What are the matter waves?	[3]
e)	Define Electric Susceptibility.	[2]
f)	Distinguish between Soft and Hard Magnetic Materials.	[3]
g)	Define interference.	[2]
h)	Explain Population Inversion.	[3]
i)	Define Time of Reverberation.	[2]
j)	What are the nanomaterials?	[3]

PART-B

(50 Marks)

2.a)	Describe seven crystal structures with diagrams.	
b)	Derive an expression for packing factor of BCC crystals.	
c)	Write notes on point defects.	[3+4+3]
	OR	
3.a)	Describe in detail the structure of Diamond.	
b)	Calculate the ratio d_{100}: d_{110}: d_{111} for simple cubic structure.	
c)	Write a note on Frenkel and Schottky Defects.	[4+3+3]
4.a)	Describe Davisson and Germer's Experiment.	
b)	Explain Maxwell-Boltzmann distribution law.	
c)	Derive an expression for effective mass of an electron.	[4+3+3]
	OR	
5.a)	Explain Heisenberg's Uncertainty Principle.	
b)	Describe Kronig-Penny Model.	
c)	Write a note on Phase space and Ensembles	[3+4+3]

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6.a) Describe the phenomenon of electronic polarization and obtain an expression for electronic polarizability.
b) Explain Domain Theory of Ferro Magnetism on the basis of Hysteresis Curve.
c) Write a note on Meissner Effect. [4+4+2]

OR

7.a) Derive an equation for Bohr Magneton.
b) Explain Classification of Dia, Para and Ferro Magnetic Materials on the basis of Magnetic Moment.
c) Distinguish between Type-I & Type-II Superconductors. [3+4+3]

8.a) Describe Newton rings experiment.
b) Explain construction of Nicol prism.
c) Derive an expression for acceptance angle and numerical aperture. [3+3+4]

OR

9.a) What is Double refraction?
b) Describe construction and working principle of Ruby laser system.
c) Write the important applications of optical fibers. [3+4+3]

10.a) What is Hall effect?
b) Describe construction and working principle of LED.
c) Describe Sol-gel method for the preparation of nanomaterials. [2+4+4]

OR

11.a) Discuss construction of Photo Diodes.
b) Derive an expression for Sabine's Formula for Reverberation Time.
c) Explain the quantum confinement effects in nanomaterials. [3+4+3]

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